CPS

Dehydration in Children

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Introduction

Newborns and young children have a much higher body water content than adolescents and adults (see Table 1) and are more prone to water, sodium (Na^+) and potassium (K^+) loss during illness.

Table 1: *Percentage of Body Water by Age Group*

Age	% Body Water
Newborn (≤1 month)	75–80
Infant (1–12 month)	70–75
Child (1–12 y)	60–70
Adolescent/adult	55–60

Goals of Therapy

- Treat shock/impending shock
- Treat dehydration using an appropriate fluid and route
- Treat electrolyte imbalances
- Prevent complications (seizures or edema)
- Provide education to family members to help prevent recurrence

Investigations

- Thorough history with attention to:
  - underlying cause(s): vomiting and/or diarrhea or other excessive fluid loss
  - frequency and amount of loss
  - frequency and amount of urinary output
- Physical examination including[1] heart rate, respiratory rate, blood pressure, temperature, oxygen saturation and capillary refill
- Laboratory tests: serum Na^+, K^+, Cl^-, urea, creatinine, glucose and bicarbonate (HCO_3^-), blood gases and urinalysis as indicated clinically

The assessment of dehydration in infants and children is challenging (see Table 2) because children are able to maintain adequate blood pressure despite moderate to severe dehydration.

Table 2: *Estimation of Dehydration*

	Severity of Dehydration		
Assessment Parameter	**Mild**	**Moderate**	**Severe**
Weight loss—Infants (<1 y)	5%	10%	15%
Weight loss—Children (≥1 y)	3–4%	6–8%	10%
History	Decreased fluid intake	Decreased fluid intake	Markedly decreased fluid intake

	Severity of Dehydration		
Assessment Parameter	**Mild**	**Moderate**	**Severe**
	Decreased urine output	Markedly decreased urine output	Anuria
	Normal activity	Listless, acute weight loss	Obtunded
			Longer duration of illness
Pulse	Normal	Slightly increased	Rapid
Blood pressure	Normal	Normal to orthostatic hypotension, >10 mm Hg change	Orthostatic hypotension to shock
Behaviour	Normal	Irritable	Hyperirritable to lethargic
Thirst	Slight	Moderate	Intense
Mucous membranes[a]	Normal	Dry	Parched
Tears	Present	Decreased	Absent, sunken eyes
Anterior fontanelle (<3 months)	Normal	Normal to sunken	Sunken
External jugular vein[b]	Visible when supine	Not visible except with supraclavicular pressure	Not visible even with supraclavicular pressure
Skin	Capillary refill <2 s	Slowed capillary refill (2–4 s), decreased	Significantly delayed capillary refill (>4 s) and

	Severity of Dehydration		
Assessment Parameter	**Mild**	**Moderate**	**Severe**
(less useful in children >2 y[a])		turgor	tenting; skin cool, acrocyanotic or mottled
Urine specific gravity	>1.020	>1.020, oliguria	Oliguria (<1 mL/kg/h) or anuria
Laboratory values	Normal urea and creatinine	Increased urea and creatinine	Significantly increased urea and creatinine, increased Hb, low glucose

[a] These signs are less prominent in patients who have hypernatremia.
[b] This may not be a reliable sign in young children.
Abbreviations Hb = hemoglobin; s = second

Therapeutic Choices

Pharmacologic Choices

Intravenous Rehydration Therapy

Treatment of dehydration involves replacing fluid deficits, then maintaining normal hydration.

Shock occurs when adequate tissue perfusion cannot be maintained. The systolic blood pressure at which this happens varies with age: newborns, <60 mm Hg; infants 1 month to 1 year, <70 mm Hg; children >1 year, <70 mm Hg + (age in years × 2). Shock often presents in children as an increased capillary refill time and an elevated heart rate (see Table 2). It must be treated aggressively using isotonic saline (NaCl 0.9%) or Ringer lactate, and in some cases albumin (see Figure 1).

The calculation of the fluid deficit for a given degree of dehydration can be based on historical or objective information, e.g., predehydration and present dehydrated weight. When the predehydration weight is known:

Deficit litres (L) = predehydration weight (kg) – present weight (kg).

For children ≥1 year, predehydration body weight can be estimated by:

Body weight (kg) = 3 × age (y) + 7. This gives an estimated weight at or about the 50th percentile for age and can be used for children up to 10 years of age.[2]

Maintenance fluid (see Table 3, Table 4) is the amount of fluid required to maintain normal hydration. Maintenance fluids are linked to caloric requirements and take into account insensible losses.

***Table 3:** Maintenance Intravenous Fluid and Electrolyte Requirements in Children*

Body Weight	Daily Fluid Requirement Method	Hourly Rate Method
≤10 kg	100 mL/kg	4 mL/kg
11–20 kg	1000 mL + 50 mL for each kg over 10 kg	40 mL/h + 2 mL/h for each kg over 10 kg
>20 kg	1500 mL + 20 mL for each kg over 20 kg	60 mL/h + 1 mL/h for each kg over 20 kg
Daily Electrolytes:		
Sodium: 2.5–3 mmol/100 mL fluid. Potassium: 2–2.5 mmol/100 mL fluid		

***Table 4:** Sample Calculation of Maintenance Intravenous Fluid and Electrolyte Requirements*

Using information for a **15 kg** child from Table 3, first estimate the fluid requirements as either an hourly or daily amount. Next, estimate basic electrolyte requirements.

	For the first 10 kg body weight	For the next 5 kg body weight	Total
Hourly Fluid Rate Method	40 mL/h	5 kg × 2 mL/kg/h = 10 mL/h	50 mL/h
Daily Fluid Rate Method	1000 mL/day	5 kg × 50 mL/kg/day = 250 mL/day	1250 mL/day

Electrolytes: (using the daily fluid values)

Na^+	1250 mL × 3 mmol/100 mL = 37.5 mmol (37.5 mmol/1250 mL or 30 mmol/L)
K^+	1250 mL × 2 mmol/100 mL = 25 mmol (25 mmol/1250 mL or 20 mmol/L) (administration of large amounts of potassium may not be tolerated, so estimate requirements using the low end of the range)

Commercially available solutions that would ensure adequate sodium replacement are NaCl 0.45% or 0.9%, with or without dextrose. Emerging evidence also supports use of Ringer lactate solution.[3]

Dehydration is classified into 3 types depending on serum Na^+ concentration (see Table 5).

***Table 5:** Types of Dehydration*

Type of Dehydration (percentage of cases)	Serum Na^+ (mmol/L)	Serum Osmolality (mOsm/kg)
Isonatremic (80%)	130–150	Normal: 280–295 mOsm/kg Equal water and salt loss
Hypernatremic (15%)	>150	Elevated: 295 mOsm/kg Water loss > salt loss
Hyponatremic (5%)	<130	Serum osmolality may be normal, elevated or reduced Must determine subgroup (see Hyponatremic dehydration)

Isonatremic dehydration (see Figure 1, Table 6) is the most common form of dehydration, with loss of both K^+ and Na^+. K^+ can be added to the IV mixture following establishment of urinary output. K^+ administration should not normally exceed 4 mmol/kg/day.[4] Higher K^+ concentrations can be used in life-threatening hypokalemia.

Hypernatremic dehydration usually develops slowly and is corrected slowly to prevent cerebral edema and seizures. Shock is treated aggressively by administering IV NaCl 0.9% until urinary output is re-established, then NaCl 0.45% + 5% dextrose in water (D5W) is used to correct dehydration states and restore Na^+ to normal levels.

The goal of therapy is to reduce serum Na^+ by 10–15 mmol/L/day and to restore hydration to normal in no less than 48 hours. If the serum concentration drops rapidly (>10–15 mmol/L/day or >1 mmol/L every 2 h), change the IV solution to NaCl 0.9% + D5W.

Hyponatremic dehydration is classified into 3 subgroups:

- Excessive water
- Na^+ depletion
- Factitious lowering of serum Na^+ concentration due to increased glucose, electrolytes, lipids and proteins

Symptomatic hyponatremia is usually related to the degree of serum Na^+ depletion. Children with serum Na^+ >120 mmol/L rarely demonstrate any clinical manifestations. When serum Na^+ drops below 120 mmol/L, neurologic manifestations (e.g., seizures) are common. Children who are symptomatic require aggressive replacement using hypertonic saline (NaCl 3%) to achieve a serum Na^+ >125 mmol/L. Hyponatremia correction should be carried out carefully and monitored closely. Correction should occur quickly until serum Na^+ is >125 mmol/L. The remaining correction above 125 mmol/L should occur over 8 hours. In general, 2–4 mL/kg of NaCl 3% is given at a rate of 1–3 mL/kg/hour. Serum sodium will increase by approximately 5 mmol/L for every 6 mL/kg of NaCl 3%.

Serum Na^+ deficit can be calculated as follows:

Na^+ deficit (mmol) = [Na^+ desired – Na^+ actual (mmol/L)] × body weight (kg) × total body water (L/kg)

After the initial elevation of Na^+ to >125 mmol/L, the remaining deficit can be replaced over 24–48 hours in asymptomatic children.

Remember that children who are dehydrated and have ongoing fluid losses need to have those fluid losses replaced. Those replacements need to be considered in addition to their estimated deficit plus maintenance fluids. The replacement of ongoing fluid losses usually occurs in a ratio of 1 mL of replacement to each 1 mL of fluid lost.[5] When determining fluid loss and its need for replacement, consider excessive loss secondary to high urinary output in a patient with diabetes, nasogastric losses, or excessive ongoing vomiting and/or diarrhea. Adjust fluid for electrolyte losses as well.[6] The replacement fluids suggested are based on the author's practice, although other fluid replacement strategies for children have been proposed.[7][8]

Table 6: Sample Calculation for Initial Management of Isonatremic Dehydration

Fluid	Total fluid replacement equals *deficit* replacement plus *maintenance*. Fluid deficit in 10% dehydration is 100 mL/kg; in 5% dehydration 50 mL/kg.
	Example: For a 15 kg child who has 10% isonatremic dehydration: Deficit replacement calculation = 15 kg × 100 mL/kg = 1500 mL Need to replace 50% or 750 mL over first 8 h at a rate of *94 mL/h* Maintenance = 50 mL/h (using calculation from Table 4) Total = *144 mL/h* (94 mL/h + 50 mL/h) for first 8 h Reduce to 100 mL/h for next 16 h (replace remaining deficit of 750 mL + maintenance over 16 h [47 mL/h + 50 mL/h ≈ 100 mL/h]). Reduce to maintenance rate as tolerated after this time.
Electrolytes	Na^+ loss would be approximately 120 mmol (8–10 mmol/kg/day) and K^+ loss would be approximately 120 mmol (8–10 mmol/kg/day).
Solution Choice	Using NaCl 0.45% + D5W at the above rates will replace the Na^+ loss in 13 h. K^+ replacement should not exceed 4 mmol/kg/day (60 mmol/day in this example). Replacement of K^+ will require 2 days.[9]

Oral Rehydration Therapy

Oral rehydration is the treatment of choice in children with mild to moderate dehydration. The WHO and the American Academy of Pediatrics (AAP) recommend oral rehydration therapy as a first-line treatment in infants and children with mild to moderate dehydration. It can be used in all types of dehydration provided that hypo- and hypernatremic dehydration are not at the extremes of the spectrum.

The fluid deficit is calculated and the rate of replacement is based upon the degree of dehydration.

In the child who is mildly to moderately dehydrated, the rate of replacement is 50 mL/kg over the first 4 hours; for the child who is moderately to severely dehydrated, the rate of replacement is 100 mL/kg over the first 4 hours. The rehydration phase may last from 4–12 hours depending upon the degree of dehydration as well as the ability of the child to tolerate oral rehydration. After the first 4 hours, replace the remainder of the deficit over the next 6–8 hours.

The fluid should be a balanced electrolyte solution acceptable to the gastrointestinal tract and should facilitate Na^+ transport. Ideal solutions for oral replacement therapy contain Na^+ 45–75 mmol/L, K^+ 20 mmol/L and glucose 20–24 g/L; 100–150 mL/kg/day is given to the child. It is impossible to determine how much fluid is being consumed during breastfeeding; breastfeeding can be continued during the process of rehydration in mild and moderate dehydration[10][11] but should not be considered the sole source of rehydration fluid. Once oral rehydration is complete, usual breastfeeding or oral feeding with formula can resume.

Commercially available preparations (see Table 7) may be used to rehydrate the child with observation in an ambulatory/emergency room setting or at home. In infants and children with mild dehydration, the type of solution used for rehydration and post-initial rehydration does not alter the outcome. In a study of children with mild dehydration, half-strength apple juice had the same rate of treatment failure as oral rehydration solution.[12]

Children who have been started on IV fluid replacement therapy should be switched to oral replacement therapy as soon as feasible since it is equally effective and is associated with a shorter hospital stay.[13] If the child is unable to tolerate oral replacement therapy secondary to vomiting, oral replacement solutions may be delivered via nasogastric tube. It is important to ensure that no contraindications (protracted vomiting despite small, frequent feeding; severe dehydration with shock-like state; impaired consciousness; paralytic ileus; monosaccharide malabsorption) are present prior to initiating oral replacement therapy.[14]

The success of oral rehydration may be improved by the use of an antiemetic. **Ondansetron** has the most evidence to support its use in this situation.[15] Compared with placebo, it has been shown to reduce the rate of hospital admission, reduce the need for IV fluids and increase the proportion of patients who stopped vomiting 1 hour after administration.[16] A single oral dose is suggested according to body weight as follows: 8–14 kg = 2 mg, 15–30 kg = 4 mg, >30 kg = 6–8 mg.[17] The most significant reported side effect is diarrhea. IV doses of 0.1–0.15 mg/kg can be administered if the patient is intolerant of the oral preparation. Oral rehydration should begin 15–30 minutes following administration of ondansetron.[17]

Table 7: Oral Rehydration Solutions

	Composition				
Product	Dextrose g/L	K^+ mmol/L	Na^+ mmol/L	Cl^- mmol/L	Cost[a]
Hydralyte Effervescent Tablets[b]	16	20	60	35	$$
Hydralyte Electrolyte Powder[b]	14.5	20	45	45	$$
Hydralyte Electrolyte Solution	16	22	46	43	$$
Pedialyte Electrolyte	25	20	45	35	$$
Pedialyte Electrolyte Freezer Pops	20	20	45	35	$$
Pedialyte Liquid	25	20	45	35	$$

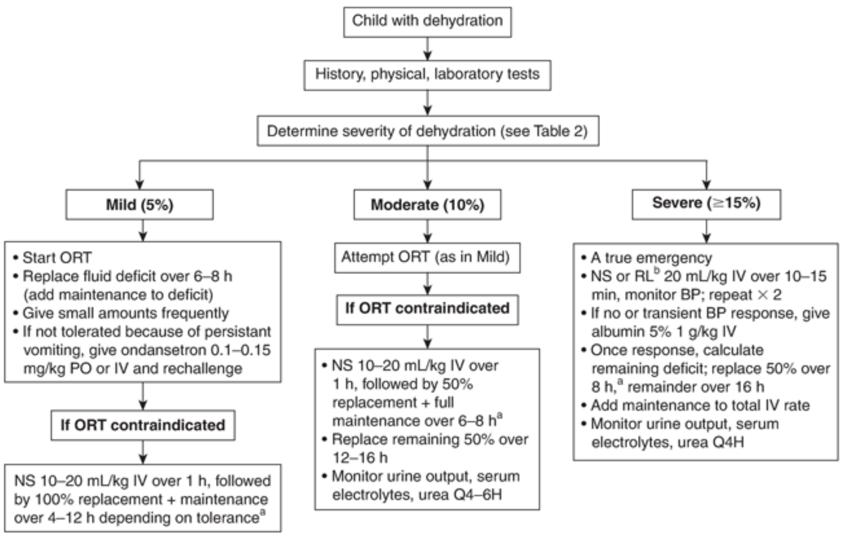
[a] Cost of smallest available pack size; includes drug cost only.
[b] Electrolyte concentration when reconstituted according to manufacturer's instructions.
Legend:
$ = < $5; $$ = $5–10

Therapeutic Tips

- Absolute indications for admission to hospital are:
  - shock
  - hypo-/hypernatremia
  - intractable vomiting/diarrhea
  - altered sensorium
- Possible indications for admission to hospital are:
  - serum HCO_3 <15 mmol/L at onset of therapy
  - poor response to oral replacement therapy and ongoing requirement for IV therapy
- Clinical signs of dehydration are often not present until at least 5% of a child's body weight is lost.
- Depending on the clinical situation, consider measuring serum electrolytes before, at the time of and after starting IV fluids.
- Consider using a rapid rehydration protocol using IV fluids in infants and children with moderate to severe dehydration.

Algorithms

Figure 1: *Management of Isonatremic Dehydration*



[a] Replacement therapy after bolus should contain at least 50–60 mmol/L Na^+ plus a source of glucose (e.g., D5W) plus appropriate K^+. Consider NaCl 0.45% (Na^+ 77 mmol/L) + D5W (5 g glucose/100 mL) + appropriate K^+. K^+ should not exceed 4 mmol/kg/day and replenishment should be done gradually over 2 days. **If no urine output, do not give K^+.**
[b] Evidence supporting the use of Ringer lactate in pediatric resuscitation is beginning to unfold.[3]

Abbreviations: BP = blood pressure; NS = normal saline, 0.9% NaCl; ORT = oral rehydration therapy; RL = Ringer lactate

Suggested Readings

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